8.4.1 Storage form of high energy compounds

Living organisms use two main types of energy storage. First one is energy-rich molecules such as glycogen and triglycerides. They store energy in the form of covalent chemical bonds. Cells synthesize such molecules and store them for later release of the energy. The second form of biological energy storage is electrochemical reactions and takes the form of gradients of charged ions across cell membranes.

8.4.2 ATP as a high energy compound

Adenosine triphosphate (ATP) is the chief carrier of energy in cells. Hydrolysis releases energy from the chemical bonds in ATP to fuel cellular processes. When energy is needed by the cell, it is converted from storage molecules into ATP. The high energy of ATP is because of its two high-energy phosphate bonds. The bonds between phosphate molecules are called phosphoanhydride bonds. Structure of ATP is represented in Figure 8.10.

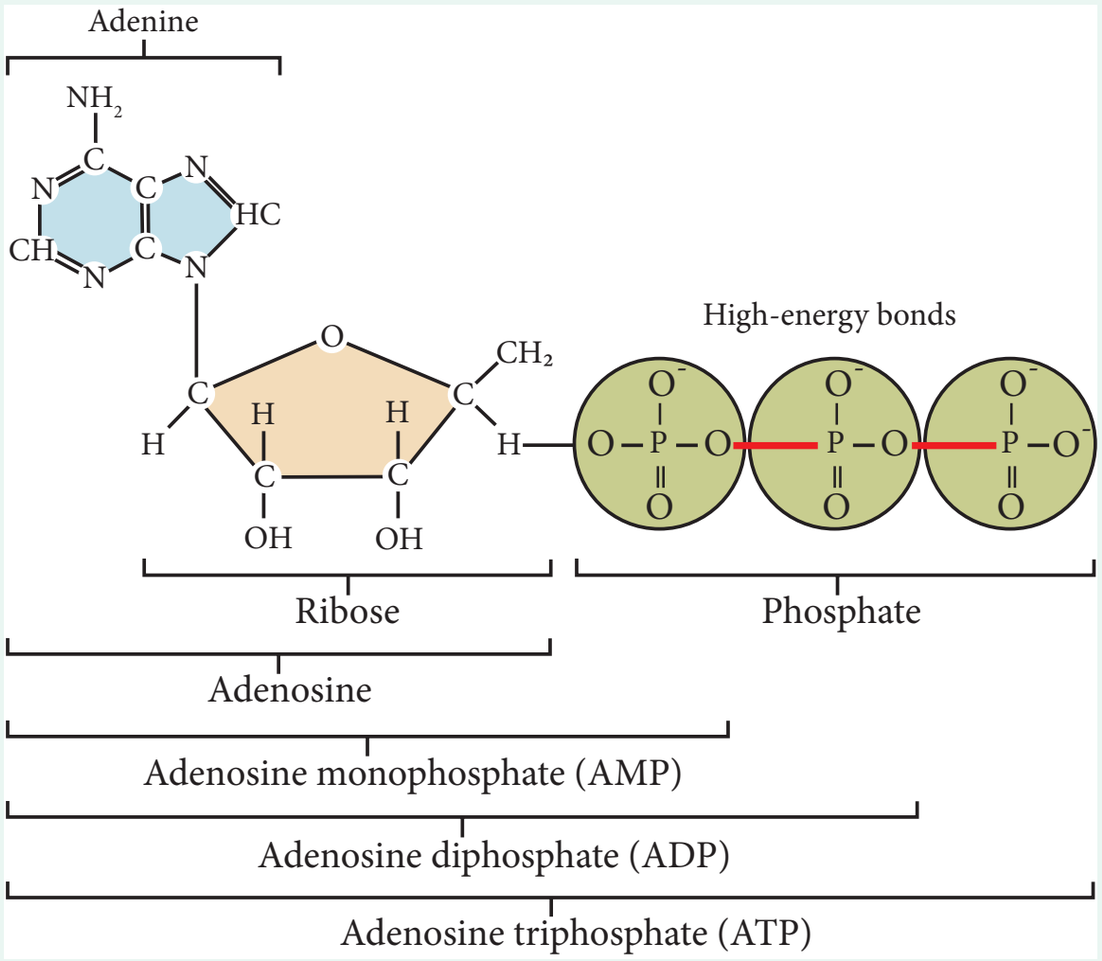

Figure 8.10 Structure of Adenosine phosphates

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